

SELENE-REG-X Scientific Lunar Image Registration and Correspondence

Context: Multi-modal, Sun-angle and scale-invariant image correspondence using Chandrayaan-2 optical imagery.

Generated: 2026-10-01T07:25:11.791972

Status: VERIFIED WITH ENVIRONMENT LIMITATION

B. Executive Summary

The implemented system provides a robust coarse-to-fine registration pipeline:

Scientific raster decoding -> Metadata/product interpretation -> Radiometric processing -> Representation generation -> Feature matching -> Geometric estimation -> Subpixel refinement -> Global graph optimization -> Mosaic generation -> Validation -> Report/provenance

STATUS: VALIDATED SYNTHETICALLY. REAL MISSION DATA NOT EXECUTED due to environment limitations.

C. System Architecture

Modules utilized: Phase 1/2 Data decoding -> Phase 3 Radiometry -> Phase 4 Scale/Pyramid -> Phase 5 Cross-Representation -> Phase 6 Geometric Estimation -> Phase 7 Subpixel Refinement -> Phase 9 Global Pose Graph -> Phase 10 Mosaic -> Phase 12 Validation.

D. Scientific Data Handling

Capabilities: Scientific raster dtype preservation, channel vs spectral distinction. PDS4 parsed recursively. Environment limit: GDAL/rasterio/ENVI are unavailable, restricting full native georeferencing extraction.

E. Radiometric / Illumination Processing

Phase 3 capability: Handles scale/offset adjustments and illumination robust transformations while preserving original scientific data for estimation.

F. Scale / GSD / Coarse-to-Fine

Phase 4 explicitly defines scale convention: $\text{scale_ratio} = \text{GSD_source} / \text{GSD_reference}$. Utilizes image-derived scale estimates in a coarse-to-fine registration schema.

G. Cross-Modal / Cross-Representation Matching

Phase 5 implementation. Deep learning descriptors (SuperPoint/LoFTR) are unavailable in this environment. Reliance on SIFT/ORB with Lowe's ratio test and spatial filtering.

H. Geometric Estimation

Phase 6 includes translation, similarity, affine, and homography models with automated RANSAC selection. Evaluates spatial coverage and inlier ratios.

I. Subpixel Refinement (SYNTHETIC VALIDATION)

Convention proven: Physical source motion (tx, ty) yields Phase 7 correction (-tx, -ty).

Taylor: 14 failures (Divergence due to ill-conditioned gradients without pyramid).

Lucas-Kanade: RMSE 0.003 px

ECC: RMSE 0.002 px

Phase Correlation: RMSE 0.082 px

Local Peak: RMSE 0.106 px

Consensus: RMSE 0.038 px, Median 0.029 px, Max 0.075 px

Consensus Success: 14 / 14 cases.

J. Match Point Inspector

Phase 8 frontend allows match filtering, overlay controls, and inlier/outlier visualization.

K. Global Multi-Image Optimization

Phase 9 convention: T_{ij} = image $i \rightarrow$ image j , T_i = root $\rightarrow$ image i . This is WEIGHTED POSE GRAPH OPTIMIZATION (Bundle adjustment = NOT IMPLEMENTED).

Ideal zero-error graphs correctly exit with ABNORMAL (gradient is exactly zero) - IDEAL BASELINE. Noisy loop correctly reduces objective and reports sub-pixel cycle consistency.

L. Scientific Mosaic

Phase 10 constructs dynamic float32 canvases. Raster sizing is computed via inclusive bounds: $\text{ceil}(\text{max}) - \text{floor}(\text{min}) + 1$. Valid area, footprints, and overlap mathematically proven. Max overlap depth successfully identified as 3 footprints.

M. Dataset Inventory / PDS4

Phase 11 recursive inventory and PDS4 parsing. Regression: 28/28 PASS.

N. Validation & Benchmarking

Phase 12 synthetically validated with 80 cases across Subpixel (14), Scale (6), Rotation (7), Geometry (4), Outlier (5), Degradation (8), Cross-rep, Graph, Mosaic, and Reject cases. Taylor divergence intentionally visible.

O. Regression Status

Phase 9: 94/94 PASS

Phase 10: 51/51 PASS

Phase 11: 28/28 PASS

Phase 12: VERIFIED WITH ENVIRONMENT LIMITATION

P. Real Mission Data Limitation

REAL MISSION DATA VALIDATION = NOT EXECUTED

Reason: No suitable actual mission dataset was available locally. Validation strictly represents synthetic performance.

Q. Environment Limitations

- PDS4/GeoTIFF metadata parsing: GDAL, ENVI, rasterio unavailable $\rightarrow$ Advanced georeferencing metadata blocked

- Cross-representation keypoints: AKAZE, SuperPoint, LoFTR unavailable -> Relies on standard SIFT/ORB matching
- Mission Scale: No local Chandrayaan-2 dataset -> Real mission data validation NOT EXECUTED

R. Reproducibility / Provenance

Validation seed: NOT AVAILABLE (Randomized synthetics fixed)

Source hash: NOT AVAILABLE

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S. Known Limitations

See Section Q and Section P for core limitations regarding mission data and GDAL extraction.

U. Runtime & Instrumentation

Execution Device: CPU

CPU: AMD64 Family 25 Model 117 Stepping 2, AuthenticAMD (6 cores)

Available RAM: 1.95 GB

Python: 3.14.3

OpenCV: 5.0.0

Capabilities:

- AKAZE: NOT AVAILABLE (Fallback: Unavailable in current environment)
- SuperPoint: NOT AVAILABLE (Fallback: SIFT/ORB)
- LoFTR: NOT AVAILABLE (Fallback: SIFT/ORB)

T. Conclusion

The system has completed implementation and synthetic/regression validation through Phase 12, with real mission-data validation remaining unexecuted because suitable local mission data is unavailable.